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Altium Schematic Instructions — EPS Board

Design Summary

This EPS board implements a solar-charged, battery-buffered power system for a 2S2P Li-Ion CubeSat stack. The architecture is:

Solar (4x face PCBs, 4S SM141K10TF each)

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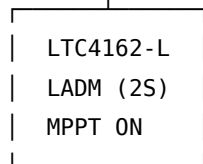
├─ X+ ──> Schottky D1 ─┐
├─ X- ──> Schottky D2 ─┐
├─ Y+ ──> Schottky D3 ─┐ SOLAR_BUS (~18V Vmp)
└─ Y- ──> Schottky D4 ─┘

```

```

|
| Impedance comp network
| (2.5 Ohm + 150uF)
|

```



```

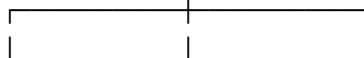
|
| VOUT (PowerPath, up to ~18V when solar present)
| VBAT (Battery Bus, 6.0–8.4V)
|

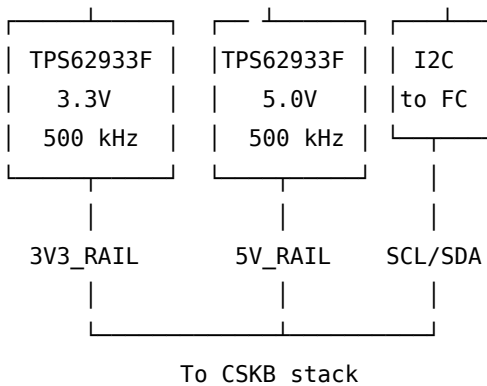
```

```

|
| VOUT_PP feeds both buck regulators
|

```





Key design decision: both buck regulators are fed directly from VOUT_PP (LTC4162 PowerPath output, 6–18V), not from VBAT. The TPS62933F’s 3.8–30 V input range covers the full PowerPath swing — including the ~18 V peak when the solar input is at MPPT — so the PowerPath topology is preserved end-to-end. This avoids the round-trip efficiency loss that would come from forcing the load through the battery, and removes the need for a separate 5 V LDO entirely.

Altium Project Setup

1. Create a new PCB project: `EPS_Board.PrjPcb`
2. Add three schematic sheets:
 - `Solar_Charger.SchDoc`
 - `Regulation.SchDoc`
 - `Connectors_Telemetry.SchDoc`
3. Each sheet uses A3 landscape format (gives more room than A4)
4. Set the project’s designator scope to “Flat” so R1, C1, etc. are unique across all sheets

Net Name Convention

Use these global net names consistently across all sheets. Use **power port symbols** (not net labels) for supply rails so they are global automatically.

Net Name	Type	Description
SOLAR_BUS	Net label	Solar input after blocking diodes
VIN_CHG	Net label	LTC4162 VIN pin, after impedance network
VOUT_PP	Net label	LTC4162 PowerPath output (pins 27, 28)
VBAT	Power port	Battery+ bus, feeds regulators
+3V3	Power port	3.3V regulated rail (global)
+5V	Power port	5.0V regulated rail (global)

Net Name	Type	Description
GND	Power port	Ground (global)
I2C_SCL	Net label	I2C clock to flight controller
I2C_SDA	Net label	I2C data to flight controller
INTVCC	Net label	LTC4162 internal 5V LDO output
VCC2P5	Net label	LTC4162 internal 2.5V LDO output
DVCC	Net label	LTC4162 I2C logic supply
SW_NODE	Net label	LTC4162 inductor switch node
SMBALERT_N	Net label	LTC4162 alert output (only discrete health signal)

Sheet 1: Solar Input & Charger

This sheet contains the solar input interface, blocking diodes, impedance compensation, and the complete LTC4162-L application circuit.

Section A: Solar Input (Top-Left of Sheet)

Text Frame (Place > Text Frame, top of this section):

SOLAR INPUT — 4 face PCBs (X+, X-, Y+, Y-), each with 4x SM141K10TF in series (~18.4V Vmp, ~23.6V Voc, ~41mA Imp per face). Faces are paralleled through blocking diodes on this board. Realistic orbit current: 40–80mA depending on attitude. Bench supply can simulate any condition through any single JST connector.

Components J1–J4: Solar Face Connectors (4x JST-PH 2-pin, B2B-PH-K-S or equivalent)

Search Manufacturer Part Search for “JST PH 2-pin” or “B2B-PH-K-S”. Place four of them in a vertical column.

Connector	Label	Pin 1	Pin 2
J1	SOLAR_XP	SOLAR_XP+	GND
J2	SOLAR_XN	SOLAR_XN+	GND
J3	SOLAR_YP	SOLAR_YP+	GND
J4	SOLAR_YN	SOLAR_YN+	GND

D1–D4: Blocking Schottky Diodes (4x)

Use a Schottky rated $\geq 30V$, $\geq 500mA$ in SOD-123 or SMA package. Suggested: SS14 (1A/40V SMA) or B5819W (1A/40V SOD-123). Pick from JLCPCB basic parts stock.

Diode	Anode	Cathode
D1	SOLAR_XP+	SOLAR_BUS
D2	SOLAR_XN+	SOLAR_BUS
D3	SOLAR_YP+	SOLAR_BUS
D4	SOLAR_YN+	SOLAR_BUS

All four cathodes connect to the common SOLAR_BUS net.

Note (Place > Note, next to the diodes):

Blocking diodes prevent shadowed/unlit face PCBs from sinking current from illuminated faces. Forward drop ~0.3V at 50mA. For bench testing, connect bench supply to any single JST connector — the diode on that input conducts and the other three are reverse-biased.

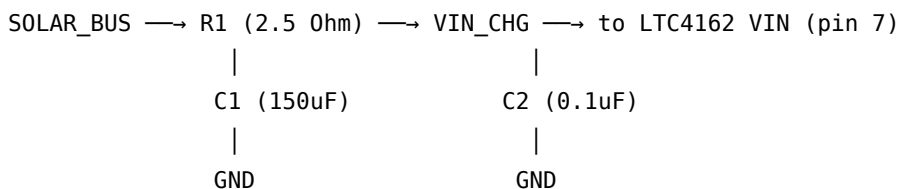
Section B: Input Impedance Compensation (Center-Left)

This network is required by the LTC4162 for solar panel MPPT stability. Without it, the high-impedance constant-current region of the solar panel can cause the MPPT control loop to oscillate. See LTC4162-L datasheet Figure 8 and the discussion on pages 33–34.

Components

Ref	Value	Package	Connection	Purpose
R1	2.5 Ohm, 1%, 1/4W	0805	Series: SOLAR_BUS → VIN_CHG	Impedance flattening
C1	150uF, 25V, Al-polymer	Radial or SMD	SOLAR_BUS to GND	Impedance comp, parallel with panel
C2	0.1uF, 50V, X7R ceramic	0402 or 0603	VIN_CHG to GND	HF bypass at LTC4162 VIN

Circuit



Note (Place > Note, next to R1/C1):

Solar panel impedance compensation per LTC4162-L datasheet Fig. 8. 2.5 Ohm + 150uF maintains ~2.5 Ohm real impedance in the 1–10kHz band where the MPPT loop operates. Required for panels with high dynamic impedance in the constant-current region. The 150uF should be solid polymer or aluminum-polymer (not wet electrolytic) for stable ESR over temperature. Panasonic ZA hybrid series recommended.

Section C: LTC4162-L and Power Path (Center of Sheet)

This is the main charger IC with its external MOSFETs, inductor, and sense resistors.

LTC4162-L Symbol Search Manufacturer Part Search for “LTC4162EUF” or place from the downloaded symbol in hardware/eps/components/LTC4162/. The IC is a 28-pin QFN (4mm x 5mm) with exposed pad (AGND, pin 29).

Place the LTC4162-L symbol in the center of the sheet with room for support components on all sides.

Pin-by-Pin Connections Input Power Path (Top-Left of IC):

Pin	Name	Connect To	Notes
7	VIN	VIN_CHG (from impedance network)	0.1uF ceramic C2 already at this node
6	INFET	MN1 gate	Input MOSFET gate drive
5	CLP	R_SNSI high side	Input current sense +
4	CLN	R_SNSI low side → VOUTA_INT	Input current sense -
3	VOUTA	VOUTA_INT, 0.1uF to GND	Analog supply output

Input MOSFET MN1 (FDMC8327L, N-channel):

```

VIN_CHG → MN1 drain
          MN1 gate ← INFET (pin 6)
          MN1 source → CLP (pin 5)
                      |
                      R_SNSI (10m Ohm, 0805)
                      |
                      CLN (pin 4) → VOUTA (pin 3)

```

Ref	Value	Package	Connection
MN1	FDMC8327L	WDFN-8 (5x3.3mm)	Drain=VIN_CHG, Gate=INFET(pin6), Source=CLP_NODE
RS1	10m Ohm, 1%, 1/2W	0805 (Kelvin)	Between CLP (pin 5) and CLN (pin 4)

Note (Place > Note, next to MN1 and RS1):

MN1: Input reverse-blocking MOSFET. LTC4162 drives gate via INFET charge pump. FDMC8327L selected per DC2038A reference design. RS1: Input current sense. $I_{IN}(MAX) = 32mV / 10mOhm = 3.2A$. Sized for bench supply flexibility — actual solar panels deliver only ~80mA. Use Kelvin connection (4-pad footprint preferred, sense traces from inner pads to CLP/CLN pins). See LTC4162-L datasheet Figure 5.

Switching Regulator (Right Side of IC):

Pin	Name	Connect To	Notes
27, 28	VOUT	VOUT_PP, 10uF ceramic to GND	PowerPath output node
25, 26	SW	SW_NODE, through L1 to CSP	Switch node
1	BOOST	22nF ceramic to SW_NODE	Bootstrap for high-side gate

VOUT (pins 27,28) —  C_VOUT (10uF, 25V, X5R) — GND
 |
 — to MN2 drain (battery MOSFET)

SW (pins 25,26) — L1 (4.7uH) — CSP (pin 21)

BOOST (pin 1) — C_BST (22nF, 10V, X7R) — SW_NODE

Ref	Value	Package	Connection
C3	10uF, 25V, X5R ceramic	0805 or 1206	VOUT_PP to GND
L1	4.7uH, >4.2A sat	Shielded power inductor	SW_NODE to CSP_NODE
C4	22nF, 10V, X7R ceramic	0402	BOOST (pin 1) to SW_NODE

Note (Place > Note, next to L1):

$L1 = 4.7\mu\text{H}$ per LTC4162-L solar application (datasheet p.49). $L = 0.3 \times V_{IN}(\text{MAX}) / f_{\text{OSC}} = 0.3 \times 23.6 / 1.5 = 4.7\mu\text{H}$. Saturation current $\geq 1.3 \times 3.2\text{A} = 4.2\text{A}$. Ferrite core required for low core loss at 1.5MHz. Coilcraft XAL5030-472MEC or equivalent.

Battery Path (Right Side of IC):

Pin	Name	Connect To	Notes
22	BATFET	MN2 gate	Battery MOSFET gate drive
21	CSP	R_SNSB high side (from L1)	Charge current sense +
20	CSN	R_SNSB low side → VBAT	Charge current sense -
19	BATSENS+	VBAT (Kelvin sense)	Battery voltage sense, 10uF ceramic

Battery MOSFET MN2 (FDMC8327L, N-channel):

VOUT_PP → MN2 drain
 MN2 gate ← BATFET (pin 22)
 MN2 source → VBAT

Charge Current Sense:

L1 output → CSP (pin 21)
 |
 RS2 (10m Ohm, 0805, Kelvin)
 |
 CSN (pin 20) → VBAT

Ref	Value	Package	Connection
MN2	FDMC8327L	WDFN-8 (5x3.3mm)	Drain=VOUT_PP, Gate=BATFET(pin22), Source=VBAT
RS2	10m Ohm, 1%, 1/2W	0805 (Kelvin)	Between CSP (pin 21) and CSN (pin 20)
C5	10uF, 10V, X5R ceramic	0805	VBAT (BATSENS+ side) to GND

Note (Place > Note, next to MN2 and RS2):

MN2: Battery reverse-blocking MOSFET. Prevents battery back-feeding the input when VIN is absent. FDMC8327L per DC2038A reference. RS2: Charge current sense.

$I_{\text{CHG}}(\text{MAX}) = 32\text{mV} / 10\text{m}\Omega = 3.2\text{A}$. With solar panels, actual charge current is panel-limited (~80mA). 10m Ω keeps headroom for bench supply testing at full 3.2A. BATSENS+ (pin 19) must Kelvin-sense to battery+ terminal, separate from the current-carrying inductor path.

Internal Regulators and Oscillator (Bottom of IC):

Pin	Name	Connect To	Notes
2	INTVCC	INTVCC, 4.7uF to GND	Internal 5V LDO, >= 6.3V rated cap
8	VCC2P5	VCC2P5, 1uF to GND	Internal 2.5V LDO, >= 4V rated cap
11	RT	63.4k resistor to GND	Sets fOSC = 1.5MHz
16	SYNC	GND	No external sync, tie to ground

Ref	Value	Package	Connection
C6	4.7uF, 10V, X5R ceramic	0805	INTVCC to GND
C7	1uF, 6.3V, X5R ceramic	0402	VCC2P5 to GND
R2	63.4k, 1%, 1/10W	0603	RT (pin 11) to GND

Note (Place > Note, next to R2):

R2 = 63.4k sets switching frequency: $f_{\text{OSC}} = 94 / R_{\text{T}}(\text{k}\Omega) = 94 / 63.4 = 1.48\text{MHz}$ (nominal 1.5MHz). Optimized for efficiency and inductor size per LTC4162-L datasheet recommendation. Place C6 and C7 as close to pins 2 and 8 as possible — these are LDO bypass caps and must return to the analog ground plane.

Cell Count Configuration (Bottom-Left of IC):

Pin	Name	Connect To	Notes
17	CELLS0	VCC2P5	2S Li-Ion config: CELLS0 = VCC2P5
18	CELLS1	INTVCC	2S Li-Ion config: CELLS1 = INTVCC

These are direct wire connections (no resistors needed). The pin strapping sets the cell count to 2 cells per the LTC4162-L datasheet Table 5.

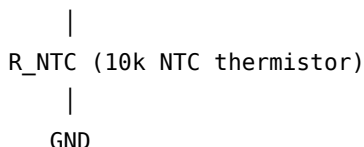
Note (Place > Note, next to CELLS0/CELLS1):

Cell count = 2S Li-Ion: CELLS1 = INTVCC, CELLS0 = VCC2P5. Matches DC2038A demo board jumper configuration for 2-cell Li-Ion. Full charge voltage = $2 \times 4.2\text{V} = 8.4\text{V}$. See LTC4162-L datasheet Table 5 (p.48) and DC2038A Table 1 (p.5).

NTC Thermistor Network (Bottom-Right of IC):

Pin	Name	Connect To	Notes
9	NTCBIAS	Through R_NTCBIAS to NTC pin	Thermistor bias output (1.2V)
10	NTC	Thermistor junction	Thermistor voltage sense input

NTCBIAS (pin 9) — R3 (10k) — NTC (pin 10)



Ref	Value	Package	Connection
R3	10k, 1%, low TC	0402	NTCBIAS (pin 9) to NTC node
RT1	10k NTC, B=3490K	0402	NTC node to GND

Suggested thermistor: Vishay NTC50402E3103FHT (10k, beta=3950K, 0402). This is the same part used on the DC2038A demo board.

Note (Place > Note, next to thermistor network):

NTC thermistor senses battery temperature for JEITA charge safety. LTC4162 applies 1.2V to NTCBIAS and reads the voltage divider on NTC. $R3 = R_{NTCBIAS} = 10k$ matches the thermistor nominal value at 25C. For higher accuracy over 0–60C, add $R_{SERIES} = 549$ Ohm in series with the thermistor and $R_{PARALLEL} = 187k$ across it (see LTC4162-L datasheet p.29, Figure 3). For prototype bring-up without a real thermistor attached, place a 10k fixed resistor in the RT1 position to simulate 25C and allow charging.

I2C and Alert (Bottom of IC):

Pin	Name	Connect To	Notes
15	DVCC	DVCC, connected to INTVCC	I2C logic level reference
13	SCL	I2C_SCL, 10k pullup to DVCC	I2C clock (open-drain)
14	SDA	I2C_SDA, 10k pullup to DVCC	I2C data (open-drain)
12	SMBALERT	SMBALERT_N, 10k pullup to DVCC	Alert output (open-drain)

Ref	Value	Package	Connection
R4	10k	0402	I2C_SCL to DVCC (pullup)
R5	10k	0402	I2C_SDA to DVCC (pullup)
R6	10k	0402	SMBALERT_N to DVCC (pullup)

Connect DVCC (pin 15) directly to INTVCC with a short trace. The datasheet requires DVCC to be the same supply as the I2C pull-up resistors.

Note (Place > Note, next to I2C pullups):

I2C address: 0x68 (7-bit). Up to 400kHz (Fast Mode). DVCC = INTVCC (~5V) sets I2C logic levels. The flight controller must be 5V-tolerant on I2C or use a level shifter. If FC runs at 3.3V I2C, either: (a) connect DVCC to 3.3V instead of INTVCC (works per datasheet if DVCC $\geq$ 2.667V), or (b) add a bidirectional I2C level shifter. Decision TBD during integration — for now, wire DVCC to INTVCC and note the open item.

Ground Pins:

Pin	Name	Connect To	Notes
23, 24	PGND	GND	Power ground, connect to copper pour
29 (exposed pad)	AGND	GND	Analog ground, thermal pad, MUST be soldered

AGND (exposed pad) must connect to PGND at a single point on the PCB. On the schematic, connect both to the GND power port.

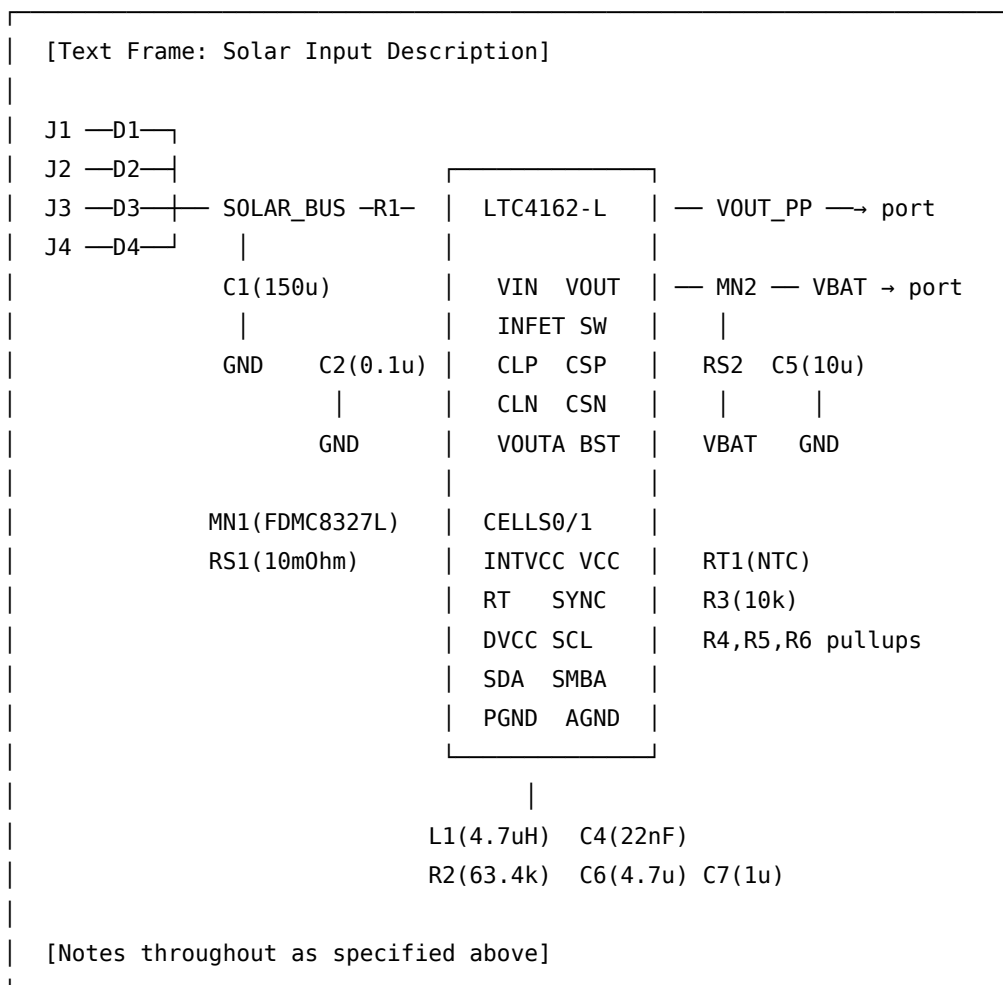
Section D: VOUT and VBAT Port Exits (Right Edge of Sheet)

Place **sheet entry ports** (or off-sheet connectors) on the right edge:

Port Name	Direction	Net	Goes To
VBAT	Output	VBAT	Sheet 3 (battery connector, test point)
VOUT_PP	Output	VOUT_PP	Sheet 2 (regulator input) and Sheet 3 (test point)
I2C_SCL	Bidirectional	I2C_SCL	Sheet 3 (CSKB connector)

Port Name	Direction	Net	Goes To
I2C_SDA	Bidirectional	I2C_SDA	Sheet 3 (CSKB connector)
SMBALERT_N	Output	SMBALERT_N	Sheet 3 (CSKB connector)
GND	Passive	GND	Global power port (auto-connects)

Sheet 1 Schematic Layout Suggestion



Sheet 1 Ports Checklist

- ☐ VBAT — battery+ bus, from CSN/BATSENS+ node

- ☐ VOUT_PP — PowerPath output, from VOUT pins 27/28
- ☐ I2C_SCL — from pin 13 with pullup
- ☐ I2C_SDA — from pin 14 with pullup
- ☐ SMBALERT_N — from pin 12 with pullup
- ☐ GND — power port (global)

Sheet 2: Regulation

This sheet takes VOUT_PP from the LTC4162 PowerPath output and produces the two regulated rails using two TPS62933F synchronous buck converters: 3.3V at up to 3A and 5.0V at up to 3A.

Text Frame (Place > Text Frame, top of sheet):

REGULATION — Both rails are produced by TPS62933F 3A synchronous buck converters fed from VOUT_PP (LTC4162 PowerPath output, 6–18V). The TPS62933F has a 3.8–30V input range, which comfortably accepts the full PowerPath swing from battery-only (~6V) to peak solar charging (~18V), preserving proper PowerPath topology.

Why TPS62933F (forced CCM variant) over the standard TPS62933: - Fixed switching frequency at all loads (no PFM transitions) - Predictable EMI spectrum — important since the comms board peer is RF-sensitive at 145 MHz and 437 MHz - Lower output ripple under light load - Cost: ~30% the price of TPSM5D1806, no LDO needed for 5V rail

Common Notes for Both Buck Stages

Both regulators are identical TPS62933F devices in SOT-583 (8-pin) packages. Place them side-by-side on the sheet so the layout mirrors the PCB placement and reviewers can compare the two stages at a glance. Datasheet references in this section are to TI document SLUSEA4D (June 2021, revised August 2022).

Pinout (TPS62933F, SOT-583):

Pin	Name	Function
1	RT	Switching frequency program (float = 500 kHz)
2	EN	Enable, with optional VIN UVLO divider
3	VIN	Power input (3.8–30 V)
4	GND	Ground
5	SW	Switch node, to inductor
6	BST	Bootstrap cap to SW (100 nF)
7	SS	Soft-start cap to GND (33 nF)
8	FB	Feedback, to top of divider

Common feedback equation: $R_{FBT} = ((V_{OUT} - 0.8) / 0.8) \times R_{FBB}$, with $R_{FBB} = 10\text{ k}\Omega$ recommended (datasheet Equation 1, §10.2.2.2).

Common switching frequency choice: 500 kHz, set by floating the RT pin (datasheet Table 9-1). Rationale: 500 kHz is the lowest standard setting and gives the best efficiency, lowest switching losses, and the cleanest fundamental away from any sensitive baseband bands. The two converters are not synchronized — the RP2040 GPIOs on the flight-controller and comms boards filter their own supply locally, so beat tones between the two buck stages are not a concern.

Section A: TPS62933F — 3.3V Rail (Left Half of Sheet)

TPS62933F Symbol — U2 (3.3V Stage) Search Manufacturer Part Search for “TPS62933F”. JLCPCB stocks the TPS62933FDRLR (SOT-583, tape and reel). Place centrally in the left half of the sheet.

Pin Connections

Pin	Name	Connect To	Notes
1	RT	Leave floating (no connect)	Selects 500 kHz
2	EN	EN UVLO divider midpoint (R20/R21)	Optional UVLO; see note
3	VIN	VOUT_PP via C30 + C31	3.8–30 V input range
4	GND	GND	Connect to thermal pad
5	SW	L20 (4.7 μ H)	Switch node
6	BST	C32 (100 nF) to SW pin	Bootstrap
7	SS	C33 (33 nF) to GND	~5 ms soft-start
8	FB	R23 / R24 divider midpoint	Output sense

Input Capacitors Per datasheet §10.2.2.9: 10 μ F X7R + 100 nF X7R, both placed directly at VIN/GND with the shortest possible loop. Voltage rating must clear the maximum VOUT_PP excursion (~18 V) with healthy derating.

Ref	Value	Package	Connection
C30	10 μ F, 35 V, X7R ceramic	1210	VOUT_PP to GND
C31	100 nF, 50 V, X7R ceramic	0402	VOUT_PP to GND

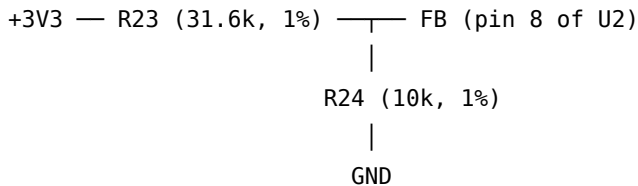
Note (Place > Note, next to C30/C31):

Input bypass per TPS62933F datasheet §10.2.2.9. C31 (100 nF) MUST be placed directly across the VIN and GND pins with the shortest possible trace — this is the high-frequency loop and is the single most layout-critical component on the buck. C30 (10 μ F) is the bulk input cap. Both rated 35 V or higher to allow the X7R DC bias derating headroom against VOUT_PP peaks of ~18 V.

Inductor Datasheet Table 10-2: 4.7 μ H at 500 kHz, 3.3 V output. Saturation current rating must comfortably exceed peak inductor current. With $K = 0.4$ ripple ratio and $I_{out} = 3$ A, peak current is approximately 3.6 A; specify a part rated ≥ 5 A saturation.

Ref	Value	Package	Notes
L20	4.7 μ H, ≥ 5 A Isat, ≤ 30 m Ω DCR	Shielded SMD (e.g., 5 $\times$ 5 mm)	Würth 74438357047 or equivalent

Feedback Divider for 3.3V



Calculation (datasheet §10.2.2.2): $R_{FBT} = ((3.3 - 0.8) / 0.8) \times 10 \text{ k}\Omega = 31.25 \text{ k}\Omega \rightarrow$ nearest E96 = **31.6 k Ω** .

Resulting VOUT = $0.8 \times (1 + 31.6 / 10) = \mathbf{3.328 \text{ V}}$ (within $\pm 1\%$ target).

Ref	Value	Package	Notes
R23	31.6 k Ω , 1%, 100 ppm/ $^{\circ}$ C	0603	Top feedback resistor
R24	10 k Ω , 1%, 100 ppm/ $^{\circ}$ C	0603	Bottom feedback resistor

Note (Place > Note, next to R23/R24):

Feedback divider for 3.3V rail. $V_{ref} = 0.8 \text{ V}$ (TPS62933F internal bandgap). $R_{FBT} = ((V_{out} - V_{ref}) / V_{ref}) \times R_{FBB}$. $R_{FBB} = 10 \text{ k}\Omega$ is the datasheet recommendation (§10.2.2.2) — lower values waste current at light load, higher values become noise- susceptible. Place R23/R24 directly at the FB pin with a short trace.

Output Capacitors Datasheet Table 10-2 calls for 40 μ F typical effective capacitance at 3.3 V / 500 kHz. Apply $\sim 2\times$ DC-bias derating for X7R 25 V parts at 3.3 V (loss is small, but conservative).

Ref	Value	Package	Connection
C34	22 μ F, 25 V, X7R ceramic	1210	+3V3 to GND
C35	22 μ F, 25 V, X7R ceramic	1210	+3V3 to GND
C36	100 nF, 16 V, X7R ceramic	0402	+3V3 to GND (HF)

Bootstrap and Soft-Start

Ref	Value	Package	Connection
C32	100 nF, 16 V, X5R ceramic	0402	BST (pin 6) to SW (pin 5)
C33	33 nF, 16 V, X7R ceramic	0402	SS (pin 7) to GND

Note (Place > Note, next to C32/C33):

C32 is the bootstrap cap — provides gate drive for the high-side FET. 100 nF X5R 16 V minimum, placed directly between BST and SW pins per datasheet §10.2.2.5. Do NOT substitute X7R 0.1 μ F in a different package; matching the example design avoids loop-area surprises.

C33 sets soft-start time. $t_{SS} = (C_{SS} \times V_{ref}) / I_{SS} = (33 \text{ nF} \times 0.8 \text{ V}) / 5.5 \mu\text{A} \approx 4.8 \text{ ms}$. Datasheet §10.2.2.4 minimum is 6.8 nF. Soft-start prevents inrush surge into the LTC4162 PowerPath when the regulator first enables.

EN / VIN UVLO Divider (Optional but Recommended) The TPS62933F can be enabled by floating EN, but adding an external UVLO divider per datasheet §9.3.6 / §10.2.2.6 prevents the regulator from attempting to start (or oscillating in current limit) when VOUT_PP collapses below a usable level — for example during a deep battery discharge before the LTC4162 latches off.

Target: VSTART = 5.5 V (rising), VSTOP = 5.0 V (falling). This sits well below the LTC4162's normal battery operating range and acts as a hard backstop.

Calculation per datasheet equations 3 and 4 ($I_p = 0.7 \mu\text{A}$, $I_h = 1.4 \mu\text{A}$, $V_{en_fall} = 1.17 \text{ V}$, $V_{en_rise} = 1.21 \text{ V}$):

- R20 (top, VIN to EN) $\approx$ **226 k Ω** (E96)
- R21 (bottom, EN to GND) $\approx$ **61.9 k Ω** (E96)

Ref	Value	Package	Connection
R20	226 k Ω , 1%	0603	VOUT_PP to EN (pin 2)
R21	61.9 k Ω , 1%	0603	EN (pin 2) to GND

Note (Place > Note, next to R20/R21):

EN UVLO divider: $V_{START} \approx 5.5$ V rising, $V_{STOP} \approx 5.0$ V falling. Backstop against deep-discharge oscillation if the LTC4162 LVD ever fails to disconnect the load. Calculated per TPS62933F datasheet §9.3.6 equations 3 and 4. R20 and R21 are SHARED between U2 and U3 (single divider, both EN pins tap the same midpoint) — the EN hysteresis currents are summed and the resulting threshold shift is well within budget for this conservative target.

Cross-reference: the EN midpoint net (R20 bottom / R21 top / U2 pin 2 / U3 pin 2) is labeled BUCK_EN and also appears on Sheet 4, where the JP_RBF Remove-Before-Flight jumper is wired across R21. Installing JP_RBF shorts R21 → forces BUCK_EN to GND → disables both bucks → kills the entire bus. See Sheet 4 Section A.

Section B: TPS62933F — 5V Rail (Right Half of Sheet)

TPS62933F Symbol — U3 (5V Stage) Identical part to U2. Place mirroring U2 on the right half of the sheet. The 5V rail feeds the comms board RF gain stages (SA612, ADL5602) and any other 5V housekeeping.

Pin Connections

Pin	Name	Connect To	Notes
1	RT	Leave floating	500 kHz
2	EN	Same midpoint as U2 (R20/R21 shared)	Common UVLO
3	VIN	VOUT_PP via C40 + C41	Same input rail as U2
4	GND	GND	Thermal pad
5	SW	L21 (6.8 μ H)	Switch node
6	BST	C42 (100 nF) to SW pin	Bootstrap
7	SS	C43 (33 nF) to GND	~5 ms soft-start
8	FB	R25 / R26 divider midpoint	Output sense

Input Capacitors

Ref	Value	Package	Connection
C40	10 μ F, 35 V, X7R ceramic	1210	VOUT_PP to GND
C41	100 nF, 50 V, X7R ceramic	0402	VOUT_PP to GND

Inductor Datasheet Table 10-2: 6.8 μ H at 500 kHz, 5 V output. Same ≥ 5 A saturation requirement.

Ref	Value	Package	Notes
L21	6.8 μ H, ≥ 5 A Isat, ≤ 35 m Ω DCR	Shielded SMD	Würth 74439346068 (datasheet reference part)

Feedback Divider for 5V Calculation: $R_{FBT} = ((5.0 - 0.8) / 0.8) \times 10 \text{ k}\Omega = 52.5 \text{ k}\Omega \rightarrow$ nearest E96 = **52.3 k Ω** .

Resulting $V_{OUT} = 0.8 \times (1 + 52.3 / 10) = \mathbf{4.984 \text{ V}}$ (within $\pm 0.4\%$).

Ref	Value	Package	Notes
R25	52.3 k Ω , 1%, 100 ppm/ $^{\circ}$ C	0603	Top feedback resistor
R26	10 k Ω , 1%, 100 ppm/ $^{\circ}$ C	0603	Bottom feedback resistor

Note (Place > Note, next to R25/R26):

Feedback divider for 5V rail. Same V_{ref} /equation as the 3.3V stage. Datasheet Table 10-2 lists $R_{top} = 52.5 \text{ k}\Omega$ for this rail; 52.3 k Ω (E96) is the closest 1% standard value and gives 4.984 V output.

Output Capacitors Datasheet Table 10-2: 20 μ F typical effective at 5 V / 500 kHz.

Ref	Value	Package	Connection
C44	22 μ F, 25 V, X7R ceramic	1210	+5V to GND
C45	22 μ F, 25 V, X7R ceramic	1210	+5V to GND
C46	100 nF, 16 V, X7R ceramic	0402	+5V to GND (HF)

Bootstrap and Soft-Start

Ref	Value	Package	Connection
C42	100 nF, 16 V, X5R ceramic	0402	BST (pin 6) to SW (pin 5)
C43	33 nF, 16 V, X7R ceramic	0402	SS (pin 7) to GND

Section C: Output Bulk Bypass and Pi-Filter Option (Bottom of Sheet)

The two 22 μ F ceramics on each rail handle the buck-stage transient response. Add board-level bulk to keep the rail quiet downstream of the regulator footprint.

Ref	Value	Package	Connection	Purpose
C50	10 μ F, 16 V, X7R	0805	+3V3 to GND	Board-level bulk, 3.3V
C51	100 nF, 16 V, X7R	0402	+3V3 to GND	HF decoupling, 3.3V
C52	10 μ F, 16 V, X7R	0805	+5V to GND	Board-level bulk, 5V
C53	100 nF, 16 V, X7R	0402	+5V to GND	HF decoupling, 5V

Note (Place > Note, next to C50–C53):

Board-level bulk bypass at the rail aggregation point. Local bypass at each consumer (RP2040, RF stages, etc.) is the responsibility of the consuming subsystem. These caps cover the board-level transient response and any trace inductance between the buck stages and the CSKB stack connectors.

Sheet 2 Port Exits (Right Edge)

Port Name	Direction	Net	Goes To
+3V3	Output	+3V3	Sheet 3 (CSKB, LEDs, test points)
+5V	Output	+5V	Sheet 3 (CSKB, test points)
GND	Passive	GND	Global power port

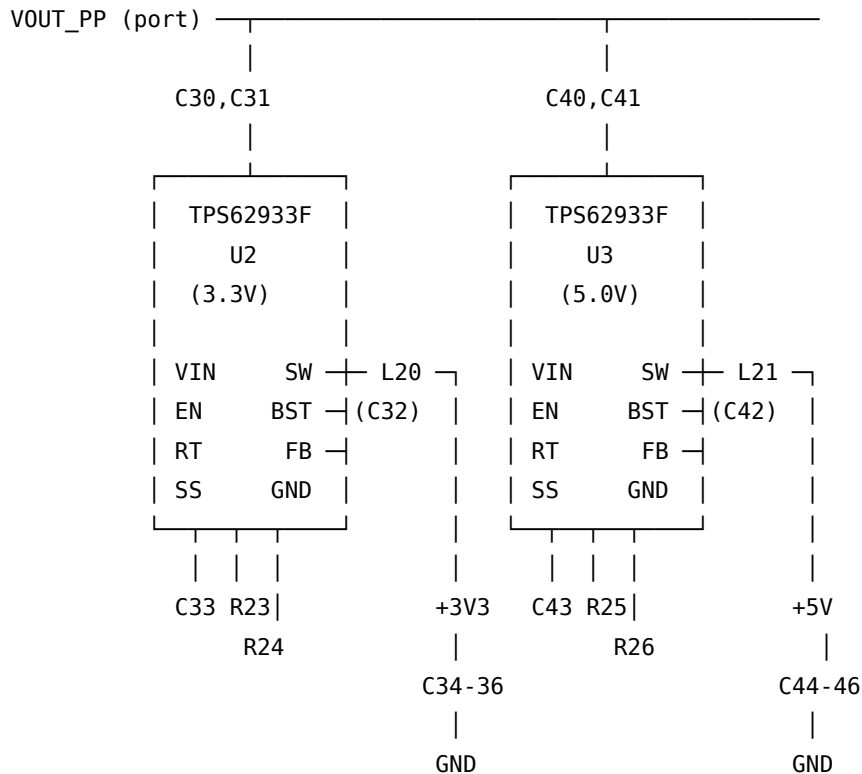
Sheet 2 Port Entries (Left Edge):

Port Name	Direction	Net	Comes From
VOUT_PP	Input	VOUT_PP	Sheet 1 (LTC4162 PowerPath output)

Note on PGOOD: the TPS62933F variant (SS pin) does not provide a PGOOD output. If a power-good telemetry signal is required for the flight controller, swap one or both stages to TPS62933P (PG pin) and route an open-drain PGOOD net to Sheet 3. For the prototype, the LTC4162's I2C telemetry already exposes input/charger health and is the primary monitoring path.

Sheet 2 Schematic Layout Suggestion

[Text Frame: Regulation Description]



EN UVLO divider (shared): R20 (VOUT_PP→EN), R21 (EN→GND)

Bulk bypass: C50/C51 on +3V3, C52/C53 on +5V

+3V3 → port

+5V → port

[Notes throughout as specified above]

Sheet 2 Ports Checklist

- ☐ VOUT_PP — input from Sheet 1 (LTC4162 PowerPath)
- ☐ +3V3 — output to Sheet 3 / global power port
- ☐ +5V — output to Sheet 3 / global power port
- ☐ GND — power port (global)

Sheet 3: Connectors & Telemetry

This sheet holds the battery connector, CSKB H1 + H2 stack connectors, status LEDs, and test points.

Text Frame (Place > Text Frame, top of sheet):

CONNECTORS & TELEMETRY — Battery interface, CSKB H1 + H2 stack connectors for inter-board power and signal distribution, status LEDs, and test points for bench validation. All subsystem power and I2C telemetry routes through the CSKB H1/H2 connectors.

Section A: Battery Connector (Top-Left)

Components

Ref	Description	Package	Connection
J5	2-pin JST-PH (battery+, battery-)	B2B-PH-K-S	Pin 1 = VBAT, Pin 2 = GND
J6	2-pin header, 2.54mm (bench supply alt)	1x2 pin header	Pin 1 = VBAT, Pin 2 = GND

Provide both a JST for the battery holder and a pin header for bench supply clip leads. They connect in parallel to the same VBAT and GND nets.

Note (Place > Note, next to battery connectors):

Battery: 2S2P LG MJ1 18650 (7000mAh total, 6.0–8.4V). J5 (JST): permanent battery holder connection. J6 (header): bench supply input for bring-up without battery. WARNING: Do not connect bench supply and battery simultaneously unless bench supply is set to match battery voltage. The LTC4162 PowerPath handles source arbitration, but bench supply on VBAT bypasses the charger path.

Section B: CSKB H1 + H2 Stack Connectors (Center of Sheet)

Connector Selection Part: 2× Samtec ESQ-126-37-G-D — 2×26 (52-pin) each, 0.1" (2.54 mm) pitch, **non-stackthrough** through-hole vertical socket. One populates the H1 (signals) position, one populates the H2 (power) position, for a total of 104 pins per Pumpkin's CSKB layout.

The EPS is the **stack endpoint** (acts as the motherboard), so it uses the non-stackthrough part (ESQ-126-37-G-D) per Pumpkin's footnote on page 17 of the Rev. E datasheet. All other boards in the centisat stack (FC, comms, payload) use the stackthrough variant ESQ-126-39-G-D.

Default stacking height is 15 mm between modules. If a module later needs taller clearance, Samtec SSQ-126-22-G-D 10 mm extensions can be inserted between specific interfaces for 24–25 mm spacing.

The mating adjacent board (FC) uses ESQ-126-39-G-D (stackthrough) on its own H1 and H2 positions.

Search Manufacturer Part Search for “ESQ-126-37-G-D” or place a generic 2×26 pin socket symbol (0.1” (2.54 mm) pitch) and assign the Samtec MPN in the component properties. **Mechanical note:** exact X/Y position of the H1 and H2 footprints on the PCB must match Pumpkin’s motherboard layout (see DS_CSK_MB_710-00484-E.pdf page 5, “Simplified Mechanical Layout”) before layout.

Pin Assignment (Pumpkin CSKB H1/H2)

Canonical reference: `system/interfaces/cskb_pinmap.md` is the authoritative pin map for the centisat stack. The table below is a board-local view and must match that file.

Update the canonical file first if anything here needs to change.

Pin numbers follow the Pumpkin CubeSat Kit Motherboard H1/H2 CSKB convention (DS_CSK_MB_710-00484-E.pdf, doc Rev. A, March 2012, pp. 13–16). All boards in the centisat stack use the same H1/H2 pin numbers for the same signal classes so that mechanical and electrical compatibility with Pumpkin / MBM2 / iSpace hardware is preserved.

Pin	Pumpkin name	centisat net	Direction (EPS)	Function
H1.41	SDA_SYS	I2C_SDA	Bidirectional	Housekeeping I2C data (LTC4162 slave 0x68)
H1.43	SCL_SYS	I2C_SCL	Bidirectional	Housekeeping I2C clock
H1.29	-RESET	SYS_RESET_N	(not driven by EPS v0.1)	Stack reset line — reserved
H1.49	USER2	EPS_ALERT_N	Output (open-drain)	LTC4162 SMBALERT_N → FC EPS_ALERT_N — EPS charger alert / fault indication
H2.25	+5V_SYS	+5V	Power Out	5 V stack rail (from TPS62933F #1)
H2.26	+5V_SYS	+5V	Power Out	5 V (parallel for current sharing)
H2.27	VCC_SYS	+3V3	Power Out	3.3 V stack rail (from TPS62933F #2)
H2.28	VCC_SYS	+3V3	Power Out	3.3 V (parallel for current sharing)
H2.29	DGND	GND	Power	Digital ground return

Pin	Pumpkin name	centisat net	Direction (EPS)	Function
H2.30	DGND	GND	Power	Digital ground return
H2.31	AGND	GND	Power	Analog ground (single plane — star-tied to DGND at EPS, single pin on CSKB)
H2.32	DGND	GND	Power	Digital ground return (third DGND pin)
H2.45	VBATT	VBAT	Power Out	Unregulated battery bus (6.0–8.4 V, monitor only on other boards)
H2.46	VBATT	VBAT	Power Out	Battery bus (parallel for current sharing)

All other H1/H2 pins are unused on the EPS board (reserved for FC SPI, UART debug, COMMS_EN/FAULT, CAN stub, and future payload signals as defined in the canonical pin map).

Note: power pins are doubled (H2.25/26, H2.27/28, H2.45/46) with adjacent DGND pins (H2.29, H2.30, H2.32). This reduces connector pin resistance and provides low-impedance current return paths. AGND is a single pin on CSKB (H2.31) and is tied to the single GND plane at the EPS board only.

Note (Place > Note, next to CSKB connectors):

CSKB stack connectors: 2× Samtec ESQ-126-37-G-D (2×26, 0.1" (2.54 mm), non-stackthrough through-hole socket, one each for H1 and H2). EPS is the stack endpoint — non-stackthrough is the correct variant per Pumpkin datasheet Rev. E page 17. Mates with ESQ-126-39-G-D (stackthrough) on the adjacent FC/comms/payload boards. Single GND plane, star-connected at the EPS — no separate analog/digital ground returns.

Bulk Bypass at Stack Connector Place bypass caps at the CSKB H2 connector for the distributed rails:

Ref	Value	Package	Connection
C60	10uF, 10V, X5R	0805	+3V3 to GND at connector
C61	100nF, 10V, X7R	0402	+3V3 to GND at connector
C62	10uF, 10V, X5R	0805	+5V to GND at connector
C63	100nF, 16V, X7R	0402	+5V to GND at connector

Section C: Status LEDs (Bottom-Left)

Ref	Color	Connection	Indicates
D10	Green	+3V3 → R30 (1k) → D10 → GND	3.3V rail active
D11	Green	+5V → R31 (1k) → D11 → GND	5.0V rail active
D12	Red	SMBALERT_N inverted or via FET	Charger alert active (optional)

D12 (optional for prototype): SMBALERT is active-low open-drain. Simplest approach: connect through a PNP inverter or skip for prototype and read SMBALERT via I2C instead.

Note: the TPS62933F (SS variant) has no PGOOD output, so the only discrete rail-health LED would have to come from a comparator on the 3.3V or 5V rail. For the prototype, the two “rail active” LEDs above are sufficient — actual fault diagnostics happen via the LTC4162 I2C telemetry path.

Ref	Value	Package
R30	1k	0402
R31	1k	0402
D10	Green LED	0603
D11	Green LED	0603
D12	Yellow LED	0603

Note (Place > Note, next to LEDs):

LED brightness at 3.3V: $I = (3.3 - 2.0) / 1k = 1.3mA$ (dim but visible). LED brightness at 5.0V: $I = (5.0 - 2.0) / 1k = 3.0mA$. For brighter LEDs, reduce resistor to 470 Ohm. For flight, remove LED population or use DNP (Do Not Populate) — LEDs waste power in orbit. Mark all LED resistors as DNP-optional in BOM.

Section D: Test Points (Bottom-Right)

In Altium, place test points as single-pin components. Search Manufacturer Part Search for “test point” or use a generic 1-pin symbol and assign a test point pad footprint (e.g., a 1.5mm round pad or Keystone 5019 loop).

How to Place Test Points in Altium

1. Place > Part > search “Test Point” in Manufacturer Part Search
2. Or: create a generic 1-pin schematic symbol called “TP”
3. Assign a footprint: use a round pad (1mm–2mm diameter) or a loop footprint (Keystone 5000-series)
4. Label each test point with its net name in the Comment field
5. On the PCB, test points appear as pads — probe them with scope/DMM

Test Point List

Ref	Net	Purpose	Measurement
TP1	SOLAR_BUS	Solar bus after diodes	Vpanel under load
TP2	VIN_CHG	LTC4162 VIN (after impedance comp)	Charger input voltage
TP3	VOUT_PP	PowerPath output	VOUT tracking behavior
TP4	VBAT	Battery bus	Battery voltage
TP5	+3V3	3.3V regulated rail	Rail voltage, ripple
TP6	+5V	5.0V regulated rail	Rail voltage, ripple
TP7	SW_NODE	LTC4162 switch node	Switching waveform, frequency
TP8	GND	Ground reference	Scope ground clip point

Place TP8 (GND) as a large pad or loop — you always need a convenient ground reference for scope probing. Consider placing two GND test points on opposite sides of the board.

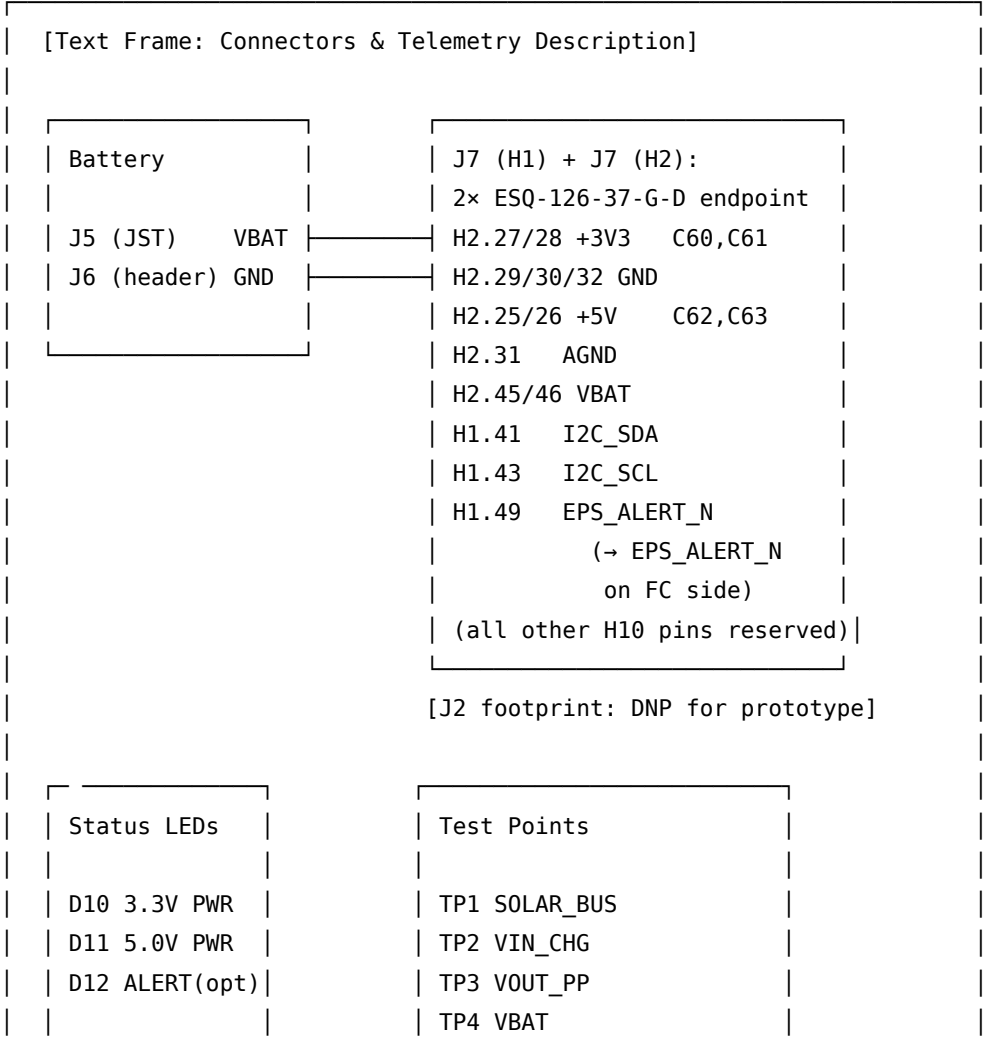
Note (Place > Note, next to test points):

All test points are single-pad footprints accessible from the top side. TP7 (SW_NODE) is critical for validating switching behavior — use a short probe lead to minimize inductance. TP8 (GND) should be a loop-style test point for easy scope ground clip attachment. For production/flight boards, test points can be reduced to TP4-TP6 (VBAT, 3V3, 5V) plus GND.

Section E: Sheet 3 Port Entries (Left Edge)

Port Name	Direction	Net	Comes From
VBAT	Input	VBAT	Sheet 1
VOUT_PP	Input	VOUT_PP	Sheet 1
+3V3	Input	+3V3	Sheet 2
+5V	Input	+5V	Sheet 2
I2C_SCL	Bidirectional	I2C_SCL	Sheet 1
I2C_SDA	Bidirectional	I2C_SDA	Sheet 1
SMBALERT_N	Input	SMBALERT_N	Sheet 1

Sheet 3 Schematic Layout Suggestion



	TP5 +3V3	
	TP6 +5V	
	TP7 SW_NODE	
	TP8 GND (x2)	
[Notes throughout as specified above]		

Sheet 3 Ports Checklist

- ☐ VBAT — from Sheet 1, to battery connector and TP4
- ☐ VOUT_PP — from Sheet 1, to TP3 (also routed to Sheet 2 regulators)
- ☐ +3V3 — from Sheet 2, to CSKB H2.27/28, LED, test point
- ☐ +5V — from Sheet 2, to CSKB H2.25/26, LED, test point
- ☐ I2C_SCL — from Sheet 1, to CSKB H1.43
- ☐ I2C_SDA — from Sheet 1, to CSKB H1.41
- ☐ EPS_ALERT_N — from Sheet 1 (LTC4162 SMBALERT_N), to CSKB H1.49 (and optional D12)

Sheet 4: Inhibits & RBF

Text Frame (Place > Text Frame, top of sheet):

EPS Sheet 4 — Inhibits & RBF (v0.1 proto)

This sheet collects all of centisat's safety interlocks in one place: the Remove-Before-Flight pin and the two deployment inhibits. On the v0.1 proto board, all three are **1x2 pin header jumpers** — a pull-pin RBF and real separation switches will replace them on a later flight spin.

Convention: **jumper installed = safe/inhibited, jumper pulled = armed/running**. Applies to all three jumpers on this sheet. The JP1 jumper on the FC board (WDT Disable) uses the same convention.

See docs/architecture/inhibit_and_deployment.md for the full policy, the polarity-flip note for real flight switches, and the list of what still needs to happen before flight.

Section A: JP_RBF — Remove-Before-Flight (Left Half of Sheet)

What it does JP_RBF is a 1x2 pin header wired **across R21** (the bottom leg of the shared EN UVLO divider on Sheet 2). When the jumper is installed, R21 is shorted out → the buck EN midpoint collapses to GND → both TPS62933F bucks (U2/U3) are disabled → +3V3 and +5V are dead → the entire satellite bus is dead. When the jumper is pulled, R21 is back in the divider and normal UVLO behavior resumes.

Important: the LTC4162 charger on Sheet 1 is **upstream** of the bucks and is **not** gated by JP_RBF. Solar → charger → battery remains alive with the RBF pin installed, so the battery can top off during pre-launch integration and on the pad. This matches CDS §3.2 intent (“charging circuit stays alive while RBF is in”).

Components

Ref	Value	Package	Connection
JP_RBF	1×2 pin header, 2.54 mm	THT	Pin 1 = buck EN midpoint (same net as R20 bottom / R21 top / U2 pin 2 / U3 pin 2); Pin 2 = GND

Placement Put JP_RBF physically adjacent to the R20/R21 UVLO divider on the PCB, and label it clearly on silk:

RBF — REMOVE BEFORE FLIGHT Installed = satellite bus OFF (charging still active).
Pulled = satellite bus ON.

Also add a bright red silkscreen border or “RBF” marker so a reviewer can see it at a glance during integration.

Note (Place > Note, next to JP_RBF):

JP_RBF — Remove Before Flight pin (v0.1 proto jumper).

Wired across R21 (Sheet 2, shared EN UVLO divider). Jumper installed = R21 shorted = buck EN forced low = +3V3 and +5V rails OFF. Jumper removed = normal UVLO operation = bus ON.

The LTC4162 charger is upstream of the bucks and is NOT affected by JP_RBF. Battery charging from solar continues while RBF is installed, so the battery can top off during integration and on the launch pad.

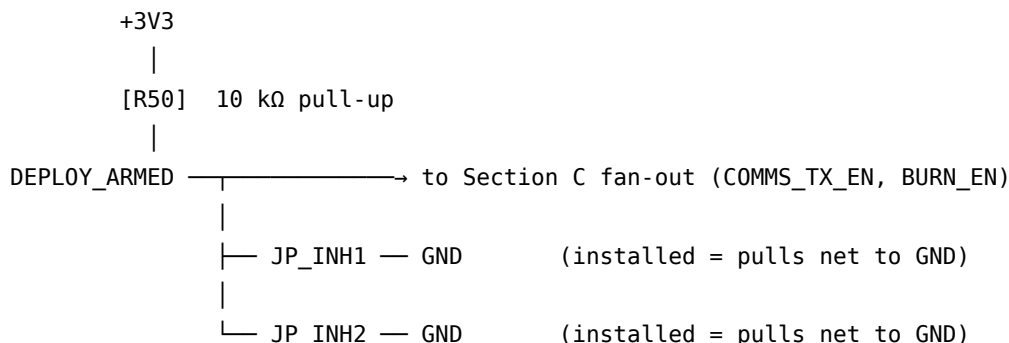
This is a proto-only implementation. The flight version will replace this jumper with a physical pull-pin through the chassis, with the same “installed = safe, pulled = armed” convention. See docs/architecture/inhibit_and_deployment.md.

Section B: JP_INH1 / JP_INH2 — Deployment Inhibits (Right Half of Sheet)

What they do JP_INH1 and JP_INH2 are **two independent** 1×2 pin header jumpers that both gate a single net, DEPLOY_ARMED. Either jumper, installed on its own, is enough to force DEPLOY_ARMED low and inhibit the satellite. Both must be removed for the satellite to arm. This gives us two independent

hardware inhibits on the proto, matching the “either switch alone keeps us inhibited” fail-safe direction of the eventual flight separation switches.

Circuit



With both jumpers removed, the 10 kΩ pull-up holds DEPLOY_ARMED high (armed). With either jumper installed, that jumper shorts the net to GND (inhibited) regardless of the other jumper’s state. The logic is: DEPLOY_ARMED = (JP_INH1 removed) AND (JP_INH2 removed).

Components

Ref	Value	Package	Connection
JP_INH1	1×2 pin header, 2.54 mm	THT	Pin 1 = DEPLOY_ARMED; Pin 2 = GND
JP_INH2	1×2 pin header, 2.54 mm	THT	Pin 1 = DEPLOY_ARMED; Pin 2 = GND
R50	10 kΩ, 1%	0402	+3V3 to DEPLOY_ARMED (pull-up)

Placement Place JP_INH1 and JP_INH2 side-by-side, clearly separated from JP_RBF so there is no confusion at integration. Silkscreen:

INH1 / INH2 — DEPLOYMENT INHIBITS Both installed = TX + deployables OFF (ground handling). Both pulled = TX + deployables ARMED. Either one alone is enough to keep the satellite inhibited.

Note (Place > Note, next to JP_INH1 / JP_INH2):

JP_INH1 / JP_INH2 — Deployment inhibit jumpers (v0.1 proto).

Both jumpers gate the DEPLOY_ARMED net in parallel. Either one installed forces DEPLOY_ARMED low via a direct short to GND, overriding R50’s pull-up. Both must be removed for the satellite to arm.

DEPLOY_ARMED fans out on this sheet to gate COMMS_TX_EN and BURN_EN (Section C). The intent is that TX and burn-wire deployables cannot activate while either jumper is present, regardless of firmware state.

Polarity note for future-you: on the flight version, these jumpers get replaced by 2× series normally-closed separation microswitches that the P-POD rails hold open while stowed. On ejection, both switches close, completing the circuit and allowing DEPLOY_ARMED to rise. Electrically that is the **inverse** of the proto jumpers (proto = short-to-GND when inhibited; flight = open-circuit when inhibited) even though the *meaning* of the net is unchanged. See docs/architecture/inhibit_and_deployment.md for the full write-up.

Section C: DEPLOY_ARMED Fan-Out (Bottom of Sheet)

DEPLOY_ARMED drives two downstream enable nets:

- **COMMS_TX_EN** — gates the comms board TX power amplifier enable. Exits Sheet 4 via a port and is carried on CSKB to the comms board. Comms TX PA is only allowed to energize when COMMS_TX_EN is high.
- **BURN_EN** — gates the antenna burn-wire deploy load switch. Burn-wire MOSFET (TBD on a future sheet or add-in board) only conducts when BURN_EN is high.

For v0.1 proto, the simplest implementation is a hard net tie: COMMS_TX_EN = DEPLOY_ARMED and BURN_EN = DEPLOY_ARMED, with the actual TX PA and burn-wire load switches living on their respective boards and accepting DEPLOY_ARMED as their hardware enable input. No AND gate, no buffer — just a labeled net exit on Sheet 4.

Future-spin improvement: add a 74LVC2G08 dual AND gate so DEPLOY_ARMED is combined with a firmware arm signal (FW_DEPLOY_ARMED) from the EPS MCU before driving COMMS_TX_EN and BURN_EN. That would add a firmware arm as the third CDS inhibit. Not in v0.1 — see the open items in docs/architecture/inhibit_and_deployment.md.

Note (Place > Note, near the DEPLOY_ARMED net label):

DEPLOY_ARMED exits this sheet to COMMS_TX_EN (CSKB to comms board) and BURN_EN (burn-wire load switch, TBD). For v0.1 proto, both downstream nets are wired directly to DEPLOY_ARMED with no intermediate gating. A future spin may insert a firmware arm via a 2-input AND gate.

Sheet 4 Ports Checklist

- ☐ +3V3 — power entry (from Sheet 2), for R50 pull-up
- ☐ GND — power entry, for jumper return paths
- ☐ DEPLOY_ARMED — internal net label (no off-sheet port — fans out on this sheet)

- COMMS_TX_EN — output, exits to Sheet 3 CSKB port entries (TBD CSKB pin, not yet allocated — see system/interfaces/cskb_pinmap.md open item)
- BURN_EN — output, exits to burn-wire driver (TBD — not on any current sheet)

Note: Sheet 4 has a **cross-sheet tie** to Sheet 2. The JP_RBF jumper pin 1 is the same net as the R20/R21 EN midpoint on Sheet 2. Use a matching net label (e.g. BUCK_EN) on both sheets so Altium's ERC ties them together on compile.

After All Four Sheets Are Done

1. Annotate

Tools > Annotate Schematic Quietly (or Annotate Schematic for manual control). Assign unique designators across all sheets: R1, R2, ... C1, C2, ... etc.

2. Compile and Validate

Project > Compile PCB Project. Check the Messages panel for: - **Net mismatch** between sheet entry ports (names must match exactly) - **Unconnected pins** — add no-connect X markers on any intentionally unconnected pins on the LTC4162 or other ICs - **Duplicate designators** — fix before proceeding - **Power port direction warnings** — these are usually benign but review

3. Fill in Title Blocks

On each sheet, double-click the title block and fill in:

Field	Sheet 1	Sheet 2	Sheet 3	Sheet 4
Title	CubeSat EPS — Solar Input & Charger	CubeSat EPS — Regulation	CubeSat EPS — Connectors & Telemetry	CubeSat EPS — Inhibits & RBF
Revision	0.2	0.2	0.2	0.2
Drawn By	Nick Grabbs	Nick Grabbs	Nick Grabbs	Nick Grabbs
Date	(today)	(today)	(today)	(today)

4. Export PDF

File > Smart PDF (or File > Print > PDF). Save to hardware/eps/releases/EPS_schematic_v0.1.pdf for review.

Complete BOM Summary

Sheet 1: Solar Input & Charger

Ref	Value	Description	Qty
J1-J4	JST-PH 2-pin	Solar face connectors	4
D1-D4	SS14 or B5819W	Blocking Schottky diodes, $\geq 30V/500mA$	4
R1	2.5 Ohm, 1%, 1/4W	Impedance comp resistor	1
C1	150uF, 25V, Al-polymer	Impedance comp cap	1
C2	0.1uF, 50V, X7R	VIN HF bypass	1
U1	LTC4162EUF-D-LADM#PBF	MPPT charger IC, QFN-28	1
MN1, MN2	FDMC8327L	N-ch MOSFET, PowerPath	2
RS1, RS2	10m Ohm, 1%, 1/2W	Current sense resistors	2
L1	4.7uH, $\geq 4.2A$ sat	Power inductor (shielded)	1
C3	10uF, 25V, X5R	VOU bypass	1
C4	22nF, 10V, X7R	BOOST cap	1
C5	10uF, 10V, X5R	BATSENS+ bypass	1
C6	4.7uF, 10V, X5R	INTVCC bypass	1
C7	1uF, 6.3V, X5R	VCC2P5 bypass	1
R2	63.4k, 1%	RT oscillator frequency	1
R3	10k, 1%, low TC	NTC bias resistor	1
RT1	10k NTC, B=3490K	Battery temperature sense	1
R4, R5, R6	10k	I2C/SMBALERT pullups	3

Sheet 2: Regulation

Ref	Value	Description	Qty
U2, U3	TPS62933FDRLR	3A sync buck, FCCM, SOT-583	2
L20	4.7 μH , $\geq 5 A$ Isat	3.3V power inductor (shielded)	1
L21	6.8 μH , $\geq 5 A$ Isat	5V power inductor (Würth 74439346068 ref)	1
C30, C40	10 μF , 35 V, X7R	Buck input bulk caps	2
C31, C41	100 nF, 50 V, X7R	Buck input HF caps (placed at VIN/GND pins)	2
C32, C42	100 nF, 16 V, X5R	Bootstrap caps	2
C33, C43	33 nF, 16 V, X7R	Soft-start caps (~5 ms)	2
C34, C35	22 μF , 25 V, X7R	3.3V output caps	2
C36	100 nF, 16 V, X7R	3.3V output HF cap	1
C44, C45	22 μF , 25 V, X7R	5V output caps	2
C46	100 nF, 16 V, X7R	5V output HF cap	1

Ref	Value	Description	Qty
C50	10 μ F, 16 V, X7R	3.3V bulk bypass	1
C51	100 nF, 16 V, X7R	3.3V HF bypass	1
C52	10 μ F, 16 V, X7R	5V bulk bypass	1
C53	100 nF, 16 V, X7R	5V HF bypass	1
R20	226 k Ω , 1%	EN UVLO top (shared U2/U3)	1
R21	61.9 k Ω , 1%	EN UVLO bottom (shared U2/U3)	1
R23	31.6 k Ω , 1%, 100 ppm	3.3V FB top resistor	1
R24	10 k Ω , 1%, 100 ppm	3.3V FB bottom resistor	1
R25	52.3 k Ω , 1%, 100 ppm	5V FB top resistor	1
R26	10 k Ω , 1%, 100 ppm	5V FB bottom resistor	1

Sheet 3: Connectors & Telemetry

Ref	Value	Description	Qty
J5	JST-PH 2-pin	Battery connector	1
J6	1x2 pin header	Bench supply alt connector	1
J7 (H1 + H2)	2 $\times$ Samtec ESQ-126-37-G-D	CSKB 2 $\times$ 26 endpoint socket, 0.1" (2.54 mm), vertical, non-stackthrough	2
D10	Green LED, 0603	3.3V power indicator	1
D11	Green LED, 0603	5.0V power indicator	1
D12	Red LED, 0603	SMBALERT indicator (optional)	1
R30, R31	1k	LED current limit	2
C60	10uF, 10V, X5R	3.3V bypass at connector	1
C61	100nF, 10V, X7R	3.3V HF bypass at connector	1
C62	10uF, 10V, X5R	5V bypass at connector	1
C63	100nF, 16V, X7R	5V HF bypass at connector	1
TP1–TP8	Test point pads	Probe points	8

Sheet 4: Inhibits & RBF

Ref	Value	Description	Qty
JP_RBF	1×2 pin header, 2.54 mm	THT	Remove-Before-Flight jumper (across R21)
JP_INH1	1×2 pin header, 2.54 mm	THT	Deployment inhibit jumper #1
JP_INH2	1×2 pin header, 2.54 mm	THT	Deployment inhibit jumper #2
R50	10 k Ω , 1%	0402	DEPLOY_ARMED pull-up to +3V3

Design Notes Summary (For Title Block or Separate Note Sheet)

These are the key design decisions documented in this schematic. Place a condensed version in the title block “Notes” field or as a text frame on Sheet 1.

1. **Battery:** 2S2P LG MJ1 18650 Li-Ion, 7.2V nom, 6.0–8.4V range
2. **Charger:** LTC4162-LADM, I2C adjustable voltage, MPPT ON by default
3. **Cell config:** 2S Li-Ion (CELLS1=INTVCC, CELLS0=VCC2P5)
4. **Solar input:** 4x face PCBs (4S SM141K10TF each), paralleled via blocking diodes. Vmp ~18.4V, Isc ~52mA per face.
5. **Regulators fed from VOUT_PP** (LTC4162 PowerPath output, 6–18V). The TPS62933F’s 3.8–30V input range covers the full range, so the PowerPath topology is preserved end-to-end (no need to bypass to VBAT).
6. **3.3V:** TPS62933F (forced CCM), 500 kHz, 31.6k/10k feedback, 4.7 μ H inductor, 2× 22 μ F X7R output
7. **5.0V:** TPS62933F (forced CCM), 500 kHz, 52.3k/10k feedback, 6.8 μ H inductor, 2× 22 μ F X7R output
8. **F variant chosen** (forced CCM, no PFM transitions) for predictable EMI spectrum next to the 145 MHz / 437 MHz comms board
9. **Shared EN UVLO divider** on both buck stages: VSTART $\approx$ 5.5 V, VSTOP $\approx$ 5.0 V — backstop against deep-discharge oscillation
10. **No PGOOD output** — TPS62933F (SS variant) has no PG pin. Health monitoring is via LTC4162 I2C telemetry. If a discrete PG signal is later required, swap to TPS62933P.
11. **Sense resistors:** 10mOhm for both input and charge — sized for 3.2A bench flexibility, actual solar current is ~80mA
12. **I2C level:** DVCC = INTVCC (~5V). FC must be 5V-tolerant or use level shifter. Open integration item.
13. **Inhibits & RBF (Sheet 4):** Three 1×2 pin header jumpers on the v0.1 proto board — JP_RBF (across R21, kills buck EN midpoint → kills bus, leaves charger alive), JP_INH1 and JP_INH2 (parallel pull-downs on DEPLOY_ARMED, either one inhibits COMMS_TX_EN + BURN_EN). Convention: installed = safe/inhibited, pulled = armed/running. Flight version replaces the in-

hibit jumpers with 2× series NC separation switches (polarity inverts, meaning unchanged) and JP_RBF with a physical pull-pin through the chassis. Full policy and breadcrumb in docs/architecture/inhibit_and_deployment.md.

Related Documents

- LTC4162-L datasheet: hardware/eps/components/LTC4162/LTC4162-L.pdf
- DC2038A demo manual: hardware/eps/components/LTC4162/DC2038A.pdf
- TPS62933F datasheet: TI SLUSEA4D (June 2021, rev August 2022)
- TPSM5D1806 datasheet (deprecated for this design): hardware/eps/components/TPSM5D1806/tpsm5d1806.pdf
- EPS architecture: hardware/eps/design/overview.md
- EPS interfaces: hardware/eps/design/interfaces.md
- Inhibit & deployment policy: docs/architecture/inhibit_and_deployment.md
- Bring-up plan: hardware/eps/bringup/phase1_validation.md
- Comms board Altium guide (reference format): hardware/comms/design/altium_digital_power_sheets.pdf